



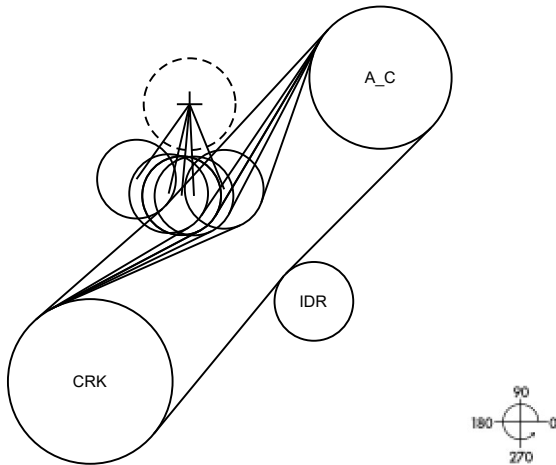
AG0902 BMC Otomotiv
Secondary Valeo TM21 AC Drive
E9843, 22Nm@27°, CCW@97.4/271.
8PK1300HD Fleetranner
30Nov2023

Belt Data, DB Ver. = 2.30.0.0, Belt Rev. = 7/14/2016

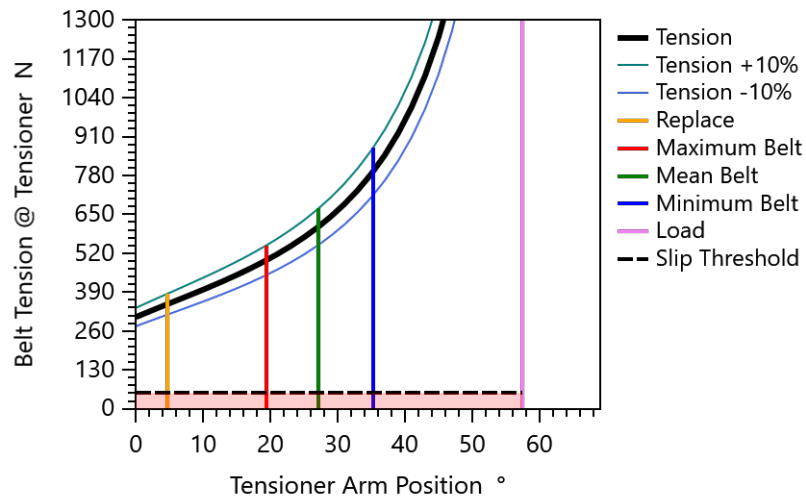
Belt Code	MT610
# of Ribs / Cord Material	8 / polyester
Effective Belt Length mm (ISO 9981)	1302

Tensioner Data

Type	Automatic
Design Tension N	608 (137 lbf)



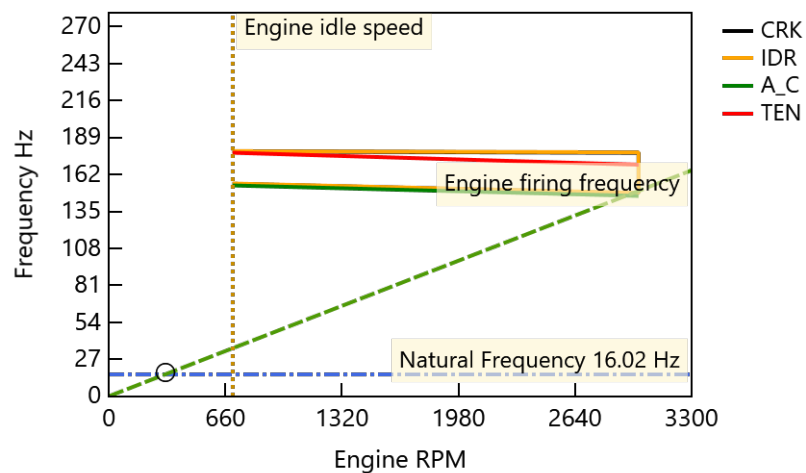
Belt Tension Control



Belt Life B10

Duty Cycle DCn-th %	Av. Vehicle Speed km/hr
95	1

Natural Frequency Map



Peak Tension & Hubload

	Engine RPM	Accel. RPM/s	Tension N	Hubload N	Direction /Wrap °
CRK	2347	1000	905	1484.4	41 / 203
IDR	2347	1000	903	84.8	319 / 5
A_C	2347	1000	611	1503.1	231 / 194
TEN	700	1000	608	330.4	134 / 31

Note: Peak Loads are defined as the highest loads (tension, shear stress and hubload) calculated for all peak load & accelerations conditions. (Not from system vibration calculation)

Permissible Grooved Pulley

Axial Offset Allowance

For 0.90° Belt Entry Angle

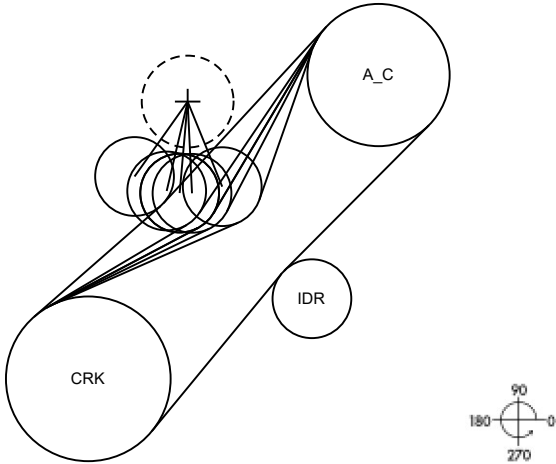
Adjacent Grooved Pulleys	Flat Pulley Angular Misalignment °	Axial Offset Allowance mm	Grooved Pulley Axial Offset %	Attributed to Flat Pulley Angular Misalignment %
A_C->CRK	IDR (0.33)	3.21	72	28
CRK->A_C	TEN (1.20)	1.1	6	94



AG0902 BMC Otomotiv
Secondary Valeo TM21 AC Drive
E9843, 22Nm@27°, CCW@97.4/271.
8PK1300HD Fleetranner
30Nov2023

Layout Data mm

	X	Y	Flat	Pitch	Effective	DOB
CRK	0.00	0.00		161.40	159.00	159.99
IDR	219.20	78.50	75.00	77.20	79.60	
A_C	284.50	297.70		139.40	137.00	137.99
TEN	89.56	182.04	75.00	77.20	79.60	



Belt Data, DB Ver. = 2.30.0.0, Belt Rev. = 7/14/2016

Belt Code	MT610
# of Ribs / Cord Material	8 / polyester
Stretch and Wear Allowance % of Length	0.70
Belt Length Tolerance (+/-) mm	5.00

Tensioner Data

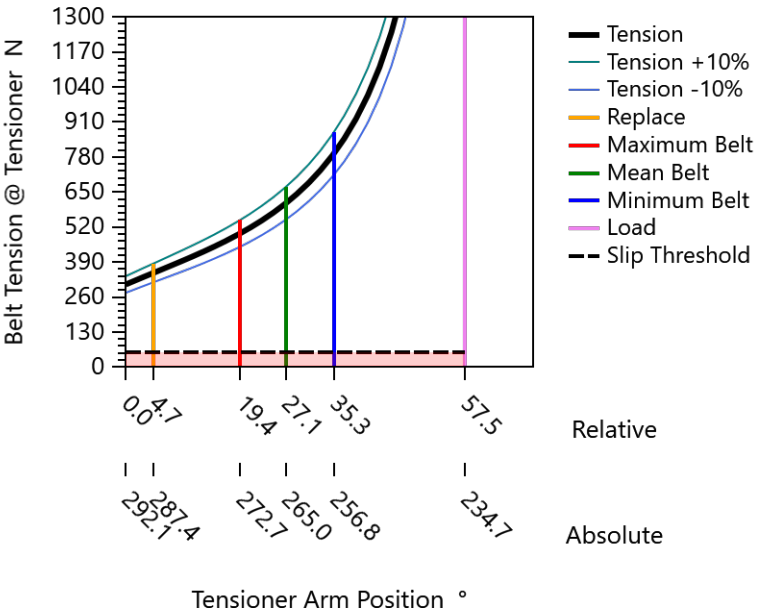
Type	Automatic
Design Tension N	608 (137 lbf)
Pivot Point {X, Y Coordinates} mm	97.40 , 271.70
Arm Length mm	90.0
Spring Mean Load Nm	22.21
Spring Pre-Load Nm	9.31
Spring Rate Nm/deg	0.476

RESULTS

Belt Drive System Geometry

	Span Length mm	Wrap Angle °	Speed Ratio (Ref. Engine)
CRK	199.9	202.7	1.000
IDR	201.5	5.4	2.091
A_C	199.1	194.1	1.158
TEN	164.1	31.4	2.091

Belt Tension Control





AG0902 BMC Otomotiv
Secondary Valeo TM21 AC Drive
E9843, 22Nm@27°, CCW@97.4/271.
8PK1300HD Fleetrunner
30Nov2023

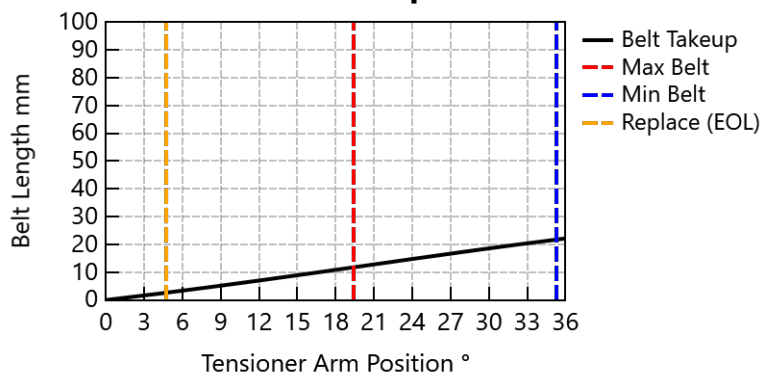
Tensioner Geometry

Belt Takeup / Tensioner Arm Ratio mm/°

0.637

Position Description	Belt-Drive Length					
	Free Arm	Replace	Max	Mean	Min	Load
Arm Position °	292.1	287.4	272.7	265.0	256.8	234.7
Idler X Coordinate mm	131.3	124.3	101.7	89.6	76.9	45.3
Idler Y Coordinate mm	188.3	185.8	181.8	182.0	184.1	198.3
Hubload to Arm Angle °	25.1	28.9	41.4	48.5	56.5	79.6
Hubload Direction °	137.2	136.3	134.1	133.5	133.3	134.2
Hubload N	243.8	265.8	311.5	329.5	348.1	414.0
Tension N	306.7	350.0	496.7	608.2	794.5	3903.9
Idler Wrap Angle °	46.8	44.6	36.5	31.4	25.3	6.1
Effective Drive Length mm	1320.3	1317.7	1308.5	1303.5	1298.5	1289.8
Required Eff. Belt Length mm	1318.5	1315.9	1306.8	1301.8	1296.8	1288.1

Belt Take-up

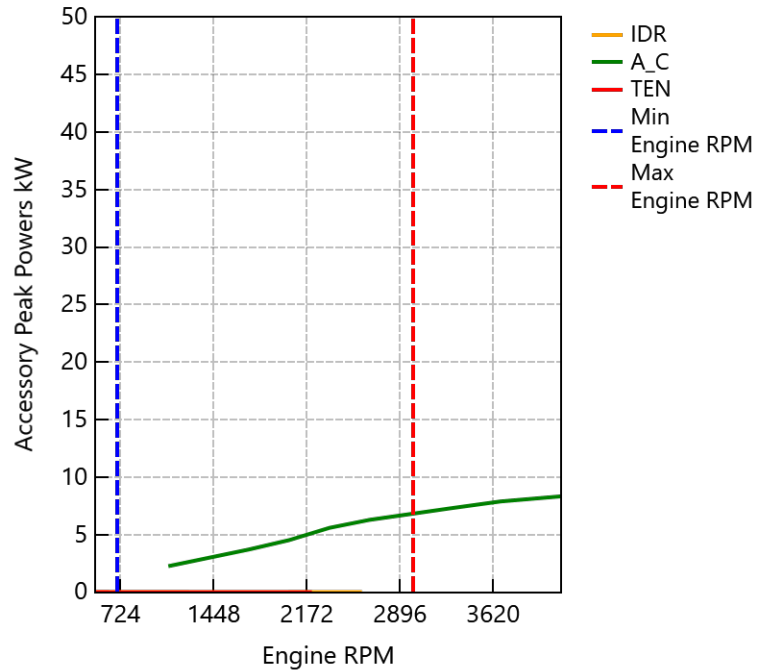




AG0902 BMC Otomotiv
Secondary Valeo TM21 AC Drive
E9843, 22Nm@27°, CCW@97.4/271.
8PK1300HD Fleetranner
30Nov2023

Design Tension N	608 (137 [lbf])
Engine Acceleration RPM/s	1000
Engine Deceleration RPM/s	1000

Accessory Peak Powers kW



Accessory Data

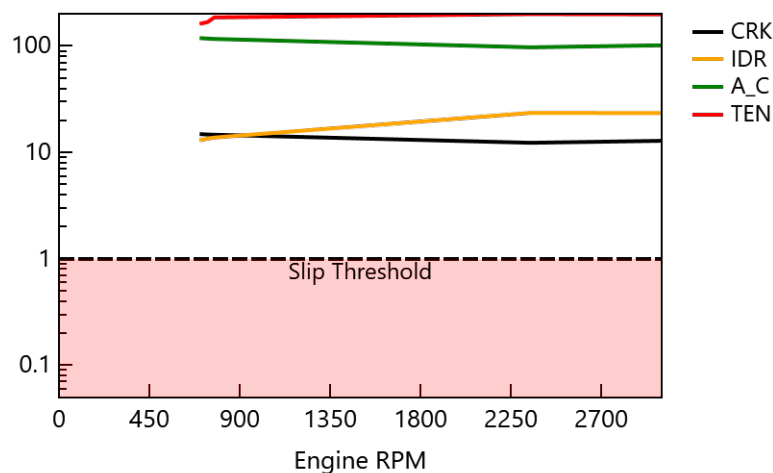
	File Name - if used	Moment of Inertia kg m ²
IDR	IDR.cmp	0.0002
A_C	Valeo-TM21-7_9kW_A_C.cmp	0.0060
TEN	TEN.cmp	0.0004

RESULTS

Belt Slip Sensitivity

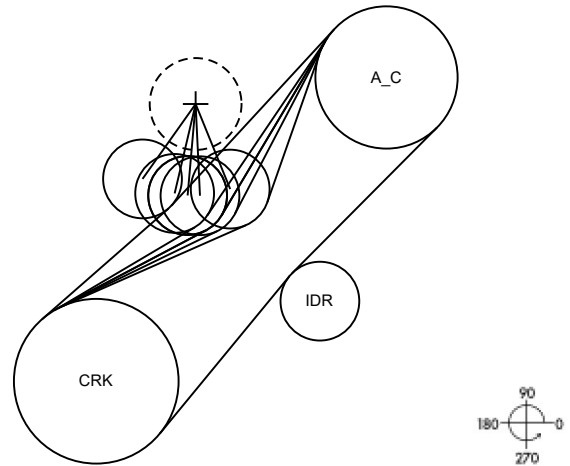
Pulley	Most Critical Load Condition					Required Increase Tension or Wrap Angle	
	Acceler. RPM/s	Load Combination %	A_C			N	°
CRK	1000	100					
IDR	1000	10					
A_C	1000	100					
TEN	1000	100					

Belt Slip Safety Factor





AG0902 BMC Otomotiv
Secondary Valeo TM21 AC Drive
E9843, 22Nm@27°, CCW@97.4/271.
8PK1300HD Fleetrunner
30Nov2023



Load Conditions kW

%	Eng. RPM	CRK	IDR	A_C	TEN
35.00	700	2.51	0.80	1.70	0.01
45.00	1200	5.01	2.00	3.00	0.01
19.00	2000	8.21	3.60	4.60	0.01
1.00	3000	11.21	4.00	7.20	0.01

Mean Hubloads N

%	Eng. RPM	CRK	IDR	A_C	TEN
35.00	700	1192	144	655	460
45.00	1200	1192	203	523	513
19.00	2000	1192	218	515	507
1.00	3000	1192	165	613	474

Mean Tensions N

%	Eng. RPM	CRK	IDR	A_C	TEN
35.00	700	608	473	186	608
45.00	1200	608	411	115	608
19.00	2000	608	395	123	608
1.00	3000	608	450	166	608